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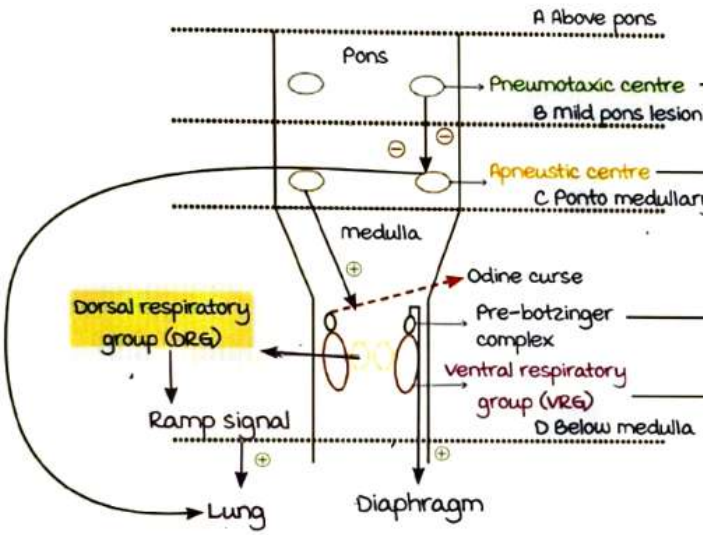
NEURAL REGULATION OF RESPIRATION

Regulation of respiration → Neural
→ Chemical: Change in $[O_a]$, $[CO_a]$, $[H+]$

Neural Regulation

00:03:57

RESPIRATORY CENTRE :



Functions
- Inhibit inspiration.
- Regulate respiratory rate (By alternating btw inspiration & expiration).

Prolong inspiration

- Pacemaker, respiratory pattern generation
  - Initiates respiration.
- maintains spontaneous breathing even during sleep.

Overdrive phenomenon

Applied aspect :

- Overdrive phenomenon : Exercise → Requires → Extra respiratory drive (VRG ↓ +) → Forceful expiration.
- Ondine curse : Absence of Pre-Botzinger complex → No Spontaneous breathing during sleep.
  - Rx : mechanical ventilation during sleep.
- Injury between medulla & spinal cord → No Spontaneous breathing (D/t disruption of impulse).
  - Seen in hanging & direct cervical injury.

RESPIRATORY PATHWAY

Initiation :

Prebotzinger complex → via phrenic nerve (C3, C4, C5) → Contracts diaphragm → Initiate inspiration.

Maintenance :

Apneustic center → ⊕ DRG → Ramp signals → Smooth sequential → Lung Stretch ↑ in tidal volume up to 500ml

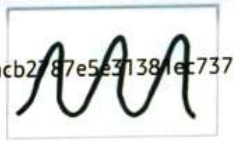
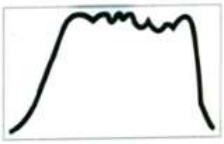
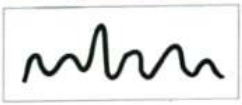
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Termination:

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- Lung stretch —Stretch reflex→ Via vagal afferent —⊖→ Apneustic center.
- Pneumotaxic center —⊖→ Apneustic center.

Waveform in lesions at respiratory centers :

Lesion		Effect	Waveform
Complete transection above pons	A	All respiratory areas intact	 Normal
Mid-pontine level section with vagus cut	B	Apneustic centre overactivity AKA apneusis	 Prolonged inspiration
Midway between pons and medulla	C	Initiation present, Regularization absent	 Irregular inspiration
Complete transection below medulla	D	Death inevitable	 Flat line

Pulmonary Reflexes

00:31:53

	Hering-Breuer inflation reflex	Hering-Breuer deflation reflex	Head's Paradoxical reflex	J Reflex
Significance	Prevent lung injury d/t to overstretching	Prevent lung collapse	To replace fluid with air in fetal lung	 Pulmonary edema
Mechanism	Prolonged inspiration ↓ Further inspiration stops ↓ Next expiration begin	Prolonged expiration ↓ Further expiration stops ↓ Next inspiration begin	Prolonged inspiration ↓ Further ↑↑ inspiration	Pulmonary edema ↓ activates J (Juxtaposition) receptors (Near blood vessels) ↓ Apnea, Breathlessness, Hypotension, Bradycardia
Neural signal	Myelinated Vagal fibres (Slow adapting)	Myelinated Vagal fibres (Slow adapting)	-	Unmyelinated Vagal fibres Slow C fibers

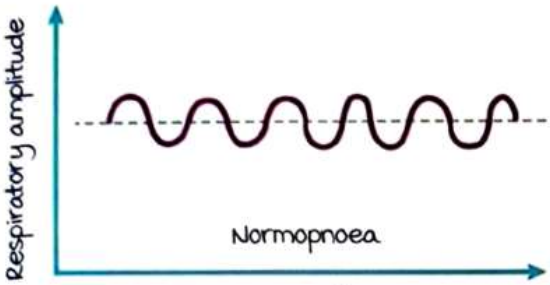
Feedback

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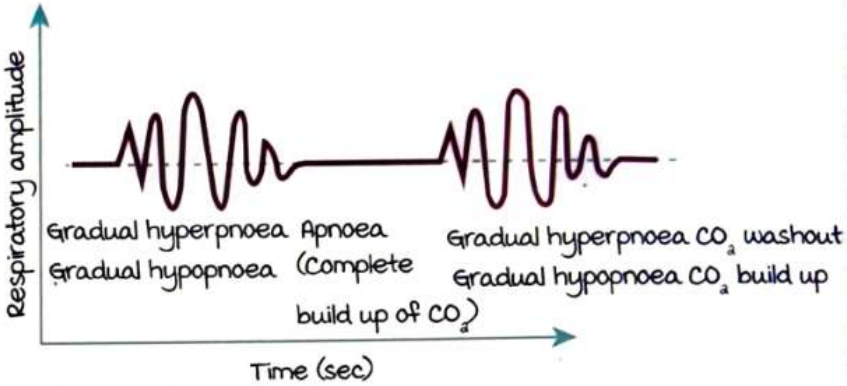
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Altered Breathing Patterns

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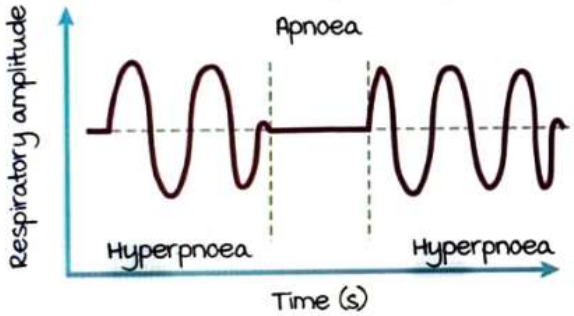
Normal respiration



Cheyne Stokes respiration

Seen in : Physiologic – sleep

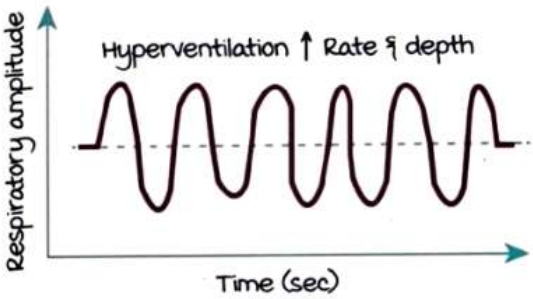
Pathologic – Congestive Cardiac failure, Uremia



Biot's [Ataxic] breathing

Seen in :

- Damage to medulla
- meningitis.



Kussmaul's Breathing/acidotic Breathing

Seen in : Diabetes Ketoacidosis

Feedback